Flight Delay / Missed Connection Risk

Choosing an Optimal Layover Buffer Under Flight Delay Uncertainty

Section 1: Introduction

When booking connecting flights, travelers face a practical decision under uncertainty: how much layover time should be left between the first flight and the connecting flight? A short layover reduces total travel time and may be more convenient, but even a moderate arrival delay on the first flight can cause a missed connection. A longer layover reduces this risk but increases waiting time and the opportunity cost of travel.

This project models the decision as a risk and cost trade-off. The key uncertainty is the arrival delay D of the first flight. Using cleaned historical U.S. flight delay data, we fit parametric delay distributions and run Monte Carlo simulation in Excel to evaluate different layover buffers. The simulation compares 45, 60, 90, and 120 minute buffers based on missed-connection probability, expected total cost, and tail-risk measures such as VaR and CVaR.

The final Excel workbook used for implementation is flight_delay_parametric_distribution_simulation.xlsx. It contains historical benchmarks, fitted distribution parameters, a zero-inflated lognormal Monte Carlo model, a zero-inflated gamma Monte Carlo model, and a fit-comparison sheet.

Section 2: Problem Formulation

We formulate the problem as a two-stage decision model. Before flight delay uncertainty is realized, the traveler chooses a layover buffer. After the arrival delay is realized, the traveler either makes or misses the connection. The realized cost is calculated scenario by scenario.

2.1 Uncertainties

The main uncertain variable is:

- D: nonnegative arrival delay of the first flight, measured in minutes.

Historical data show that arrival delay is zero-inflated and right-skewed. In the cleaned dataset, P(D = 0) = 0.6104, while P(D > 0) = 0.3896. Because the positive-delay distribution has a long right tail, we model D using two fitted alternatives:

- Zero-inflated lognormal: D = 0 with probability 0.6104; otherwise ln(D) ~ Normal(2.7697, 1.3197^2).
- Zero-inflated gamma: D = 0 with probability 0.6104; otherwise D ~ Gamma(alpha = 0.3826, beta = 92.4003).

Both distributions are simulated with 10,000 Monte Carlo trials in Excel. The lognormal model captures right-skewed delay behavior, while the gamma model provides a second fitted distribution for sensitivity analysis.

2.2 Upfront Decision Variables

The first-stage decision is made before the arrival delay is known:

- L: selected layover buffer in minutes, where L is in {45, 60, 90, 120}.

The model also uses the following cost and risk-preference parameters from the Excel Inputs sheet:

- TicketCost = $350
- ValueOfTime = $40 per hour
- MissedConnectionCost = $650
- Risk aversion parameter lambda = 0.5
- Risk tolerance alpha = 0.05, corresponding to a 95% service level target.

2.3 Wait-and-See Decision Variables

After D is realized, the wait-and-see outcome is:

- M_L(D) = 1 if D > L, and 0 otherwise. This indicates whether the traveler misses the connection.
- RealizedMissedCost_L = M_L(D) x MissedConnectionCost.
- TotalCost_L = TicketCost + (L / 60) x ValueOfTime + RealizedMissedCost_L.

2.4 Objective Function

The model chooses the layover buffer that balances expected cost and tail risk. For each candidate buffer L, the risk-adjusted objective is:

min_L E[TotalCost_L] + lambda x CVaR_95(TotalCost_L)

Expected cost captures the average cost of the itinerary, while CVaR_95 captures the average cost in the worst 5% of outcomes. This is important because missed connections are relatively rare but costly.

2.5 Constraints

- Layover choice constraint: L is in {45, 60, 90, 120}.
- Service-level constraint: P(D > L) <= alpha = 0.05.
- Budget feasibility constraint: TicketCost <= Budget. In the base case, TicketCost is fixed at $350.
- Scenario logic constraint: M_L(D) = 1 only when D exceeds the selected layover buffer.

Section 3: Solution and Recommendation

The simulation uses 1,645,408 cleaned historical observations to estimate the delay distribution. The historical data provide the following benchmarks:

Metric	Historical Value
P(D = 0)	61.04%
Mean nonnegative delay	13.77 minutes
P95 delay	74 minutes
P99 delay	176 minutes
P(D > 45)	8.81%
P(D > 60)	6.45%
P(D > 90)	3.71%
P(D > 120)	2.27%

The fitted distributions are close to the historical benchmark in the parts of the distribution that matter most for missed-connection risk. The lognormal simulation produces P(D > 90) = 3.93%, while the gamma simulation produces P(D > 90) = 4.62%. Both are close to the historical value of 3.71%.

Distribution	P(D = 0)	Mean Delay	P95 Delay	P99 Delay	P(D > 90)
Historical	61.04%	13.77	74.00	176.00	3.71%
Zero-inflated lognormal	60.66%	15.61	72.89	208.49	3.93%
Zero-inflated gamma	61.65%	13.45	83.27	194.36	4.62%

Using the base-case cost assumptions from Excel, the simulated results for the zero-inflated lognormal model are:

Layover Buffer	Missed Connection Probability	Expected Total Cost	CVaR 95	Risk-Adjusted Cost
45 minutes	8.74%	$436.81	$1,030.00	$951.81
60 minutes	6.41%	$431.67	$1,040.00	$951.67
90 minutes	3.93%	$435.55	$920.90	$896.00
120 minutes	2.53%	$446.45	$758.90	$825.90

The gamma model gives a similar qualitative conclusion:

Layover Buffer	Missed Connection Probability	Expected Total Cost	CVaR 95	Risk-Adjusted Cost
45 minutes	9.64%	$442.66	$1,030.00	$957.66
60 minutes	7.38%	$437.97	$1,040.00	$957.97
90 minutes	4.62%	$440.03	$1,010.60	$945.33
120 minutes	2.95%	$449.18	$813.50	$855.93

The 45- and 60-minute buffers have the lowest time cost, but their missed-connection probabilities exceed the 5% risk tolerance. The 90-minute buffer is the minimum buffer that satisfies the service-level constraint under both fitted distributions. It reduces missed-connection probability below 5% while avoiding the extra waiting time of a 120-minute layover.

Our primary recommendation is therefore a 90-minute layover buffer for a typical traveler. For a highly risk-averse traveler or a traveler facing a hard deadline, the 120-minute buffer is preferred because it has the lowest CVaR and the lowest risk-adjusted cost in both fitted models. In short, 90 minutes is the best practical minimum buffer, while 120 minutes is the conservative risk-minimizing buffer.

Section 4: Discussion and Future Work

4.1 Model Design Trade-off

The model balances realism and tractability. A fully realistic flight-connection model would include airline schedules, airport layout, boarding cutoffs, ticket prices, passenger type, cancellation risk, and rebooking options. To keep the model transparent and implementable in Excel, we focus on the most important driver of missed-connection risk: the arrival delay of the first flight.

The zero-inflated distribution structure is important because the empirical delay data contain many zero-delay observations and a long right tail. Modeling the positive delays with lognormal and gamma distributions allows us to test sensitivity to distributional assumptions while still keeping the simulation understandable.

4.2 Limitations

- Ticket cost is assumed to be fixed at $350 in the base model. Real itineraries would have different prices for different layover options.
- Missed connection cost is fixed at $650. In reality, this cost could also be random and depend on hotel needs, rebooking availability, or missed activities.
- The model focuses on arrival delay and does not explicitly model cancellations, diversions, gate distance, minimum connection time, or passenger-specific mobility.
- The simulation uses an aggregate delay distribution. A more detailed version could condition the distribution on airline, airport, month, day of week, and departure hour.
- The fitted distributions approximate the historical delay tail but do not perfectly reproduce rare extreme events.

4.3 Future Work

- Add itinerary-specific ticket prices and compare real route options rather than generic buffers.
- Model missed-connection cost as a distribution rather than a fixed value.
- Condition delay distributions on airline, origin airport, destination airport, month, day of week, and departure hour.
- Include cancellation and diversion scenarios with separate recovery costs.
- Use actual minimum connection times and airport terminal transfer times to make the missed-connection rule more realistic.

Signature Section

The team divided the work as follows:

- Ruiyu Zhao: project framing and presentation slides
- Yuqi Zhang: final report integration
- Zhihao Feng: data review, distribution-fitting support, and final report integration
- Binbo Xu: Excel Monte Carlo workbook, parametric simulation implementation
- Yutong Gao: project framing and presentation slides
- Yu (Richard) Fu: final report integration.